

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B.Tech. Branch: Mechanical Engg./Mechanical Engg. (Sandwich) Semester : VII

Subject Code & Name: BTMEC701 Mechatronics

Max Marks: 60

Date: 02.01.2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) What are the different elements of a Mechatronics system? Explain it with an example.	CO1	6
B) What is the difference between a sensor and transducer? How a tactile sensor works?	CO1	6
C) How a capacitive level sensor works? Explain it with a sketch.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Why signal conditioning is done? Explain the steps involved in it.	CO2	6
B) Explain the working principle of seven segment displays? Give applications of seven segment displays.	CO2	6
C) Why Analog to Digital Conversion (ADC) is required? Explain it with a neat sketch.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) How a stepper motor works? Explain it with neat sketch.	CO3	6
B) How sequencing of pneumatic cylinders is made using cascade method? Explain it with an example.	CO3	6
C) How a solenoid works? Explain it with an application using an electro-pneumatic circuit.	CO3	6
Q.4 Solve Any Two of the following.		12
A) Draw and explain the architecture of the 8085 microprocessor.	CO4	6
B) What are the addressing modes of the 8085 microprocessor?	CO5	6
C) What do you mean by microcontroller? Differentiate between microcontroller and microprocessor.	CO5	6

Q. 5	Solve Any Two of the following.		12
A)	Differentiate between Open loop system and closed loop system giving suitable examples.	CO5	6
B)	Define transfer function. Explain procedure to determine transfer function of a control system.	CO5	6
C)	What is a PID Controller? How Does a PID Controller Work?	CO6	6

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